

Skill Practice Exercise : JOINS, UNION, Filtering and Aggregates.

Question 1 : INNER JOIN

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

Question 1.2: LEFT JOIN

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

Question 1.3: RIGHT JOIN

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

Question 1.4: FULL OUTER JOIN

EmployeeID	FirstName	Last Name	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100 000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75 000	Pending Ongoing
2	Alice	Smith	HR	60000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80 000	Pending

3. Filtering Statements

7.

EmployeeID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

8.

EmployeeID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

9.

ProjectID	Project Name	Budget	Status
102	Employee Training	50 000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

10.

ProjectID	Project Name	Budget	Status
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

11.

EmployeeID	First Name	Last Name	Department	Salary	City
6	Michael	Clark	Finance	80 000	New York
3	Bob	Johnson	Finance	75 000	Toronto
1	John	Doe	IT	70 000	New York

12.	EmployeeID	FirstName	LastName	Department	Salary
	6	Michael	Clark	Finance	80 000
	3	Bob	Johnson	Finance	75000
	4	David	Brown	IT	72000

4. Aggregate Functions with Group BY & HAVING

13.	TotalSalary	Department
	155 000	Finance
	142 000	IT
	65000	Sales
	60 000	HR

14.	AverageSalary	City
	75 000	New York
	75 000	Toronto
	72 000	London

15.	EmployeeCount	Department
	2	IT
	2	Finance

16.	ProjectCount	Status
	2	Completed
	2	Ongoing
	2	Pending

17.	TotalProjectBudget	FirstName	LastName
	175 000	John	Doe